

# STUKACH

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USER GUIDE

BLENDER 5.2 - V1.8.0 · MAYA 2025 - V1.3.0

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STUKACH checks a scene for typical pipeline errors: topology, transforms, UVs, naming, materials and hierarchy structure. The artist sees the defects right in the viewport and fixes them before delivery, while the supervisor gets a formal verdict in one click. This document covers both versions: the main part describes the Blender version, section 11 covers Maya.

*Blender: [github.com/abyrvalg379/STUKACH](https://github.com/abyrvalg379/STUKACH) · Maya: [github.com/abyrvalg379/STUKACH\\_Maya](https://github.com/abyrvalg379/STUKACH_Maya)*

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# 1. ABOUT STUKACH

## 1.1 What STUKACH does

STUKACH is an asset quality checker embedded in a DCC panel. It replaces the "manual inspection" of an asset before delivery with formalized validation: a single run checks the scene objects against dozens of rules, highlights defects right on the viewport geometry, fixes typical problems with one button, and produces a report readable by both the artist and the supervisor.

Key features:

- 42 checks in 7 categories + scene units check (Blender); 43 checks + two scene checks (Maya);
- asset hierarchy validator - 13 structure rules (roots, layers, groups, branch naming);
- GPU overlays: defects are painted in color right on the geometry - faces, edges, vertices;
- Live mode: validation recalculates itself on edits, no re-RUN needed;
- one-button fixes: transforms, hierarchy, mesh data naming, garbage cleanup;
- Coordinator Mode: blockers and warnings only, a formal verdict;
- reports: clipboard copy, JSON/CSV/HTML, pre-flight before FBX/USD export;
- check set presets shared with the team as files.

## 1.2 Philosophy: three severity levels

Every STUKACH finding has a severity level. It is not a checkbox importance - the level directly defines the impact on the asset status and who makes the decision.

| LEVEL          | MEANING                                                                                                                                                   | STATUS IMPACT                       |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| <b>BLOCKER</b> | The asset cannot be delivered. The problem is guaranteed to break rendering, export, simulation or the downstream pipeline. Must be fixed before handoff. | Drives the asset status to CRITICAL |
| <b>WARNING</b> | A real problem, not always catastrophic. The artist must consciously decide - fix it or justify an exception.                                             | Drives the status to REVIEW         |
| <b>INFO</b>    | Information for review. Not an error - it highlights spots requiring a conscious decision (triangles, poles, open edges).                                 | No status impact                    |

## 1.3 Two modes: Artist and Coordinator

The panel works in one of two modes. Artist Mode is the artist working mode: the full check set, viewport overlays and Fix buttons. Coordinator Mode is the acceptance mode: only BLOCKER and WARNING are shown, with the asset verdict and a Copy Report button. The Coordinator Lock switch (Preferences) additionally hides every Fix button in Coordinator Mode - the coordinator validates only. See section 10 for working in each mode.

## 1.4 Two versions

|            | BLENDER VERSION                | MAYA VERSION                                     |
|------------|--------------------------------|--------------------------------------------------|
| Version    | v1.8.0                         | v1.3.0                                           |
| Package    | Blender 5.1+ extension (zip)   | Maya 2025 Win x64 (python + C++ overlay plugin)  |
| Panel      | View3D N-panel - STUKACH tab   | Docked panel to the left of the Attribute Editor |
| Checks     | 42 + Scene Units               | 43 + Scene Units + Empty Groups                  |
| Hierarchy  | 13-rule validator + fixes      | Built with locators; validated in Blender        |
| Repository | github.com/abyrvalg379/STUKACH | github.com/abyrvalg379/STUKACH_Maya              |

*The Blender and Maya version numbers are independent. The Maya version covers validation at the modeling stage, the Blender version - at the finished asset acceptance stage.*

## 2. INSTALLATION

### 2.1 Blender 5.1+

- Download the zip of the latest release: [github.com/abyrvalg379/STUKACH - Releases](https://github.com/abyrvalg379/STUKACH/releases) - the STUKACH.zip asset.
- Blender - Edit - Preferences - Get Extensions - the arrow menu in the top right corner - Install from Disk - pick the zip. Drag-and-drop installation works as well.
- Make sure the STUKACH extension is enabled (the checkbox).
- The panel appears in the 3D viewport N-panel (N key) - STUKACH tab.
- Updating: download the new zip and install it the same way, on top of the old version. Settings and presets survive.

*After the first launch it is recommended to open Preferences - Add-ons - STUKACH and press Save Preferences to pin the extension state.*

### 2.2 Maya 2025 (Windows x64)

- Download the kit of the latest release: [github.com/abyrvalg379/STUKACH\\_Maya - Releases](https://github.com/abyrvalg379/STUKACH_Maya/releases) - STUKACH\_Maya\_vX.Y.Z.zip.
- Unpack it. Drag the install\_stukach.py file with the mouse straight into the viewport of an open Maya - that is the stock drag-and-drop installer.
- The installer copies the package into Documents/maya/2025/scripts, the overlay plugin into Documents/maya/2025/plugin-ins, and creates a STUKACH button on the shelf (Custom tab).
- Restart Maya (or load the plugin manually: Windows - Settings/Preferences - Plug-in Manager - stukachDrawOverride.mll).
- The panel opens with the STUKACH shelf button. By default it docks to the left of the Attribute Editor.

*Dependencies: numpy and scipy (installed into the Maya user site-packages). The binary overlay (stukachDrawOverride.mll) is built for Maya 2025 Windows x64 - on other OS/versions the panel works and the overlay falls back to display layers.*

## 3. QUICK START (BLENDER)

Five steps to the first asset report:

- Open the N-panel (N) - the STUKACH tab - press RUN STUKACH. By default the selection is validated (Selected scope); press Scene for the whole scene.
- Wait for the progress (heavy scenes are validated in batches, the panel shows "Validating N/M"). The Objects list shows only objects with problems, worst on top.
- Click an object in the list - its per-check findings unfold. The Sel button selects the defective geometry and frames the camera; Show only highlights it in the viewport.
- Fix: some checks carry a Fix button (unapplied transforms, mesh data naming, garbage cleanup, hierarchy fixes). The rest is edited by hand - the overlay shows exactly where the defect is.
- Deliver: Copy Summary puts a report into the clipboard; in Coordinator Mode - Copy Report with the formal READY / REVIEW / BLOCKED verdict.

*You can now enable Live - validation recalculates itself on geometry, UV and transform edits.*

## 4. BLENDER VERSION INTERFACE

The N-panel runs top to bottom: run and modes - score and navigation - validation scope - checks - hierarchy - objects - status and export. Each block below.

### 4.1 Run and modes

| ELEMENT                                 | WHAT IT DOES                                                                                                                                                                                 |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>RUN STUKACH / STUKACH ACTIVE</b>     | Start and stop of validation. The first RUN validates the selection (Selected); widen the scope with the Scene / Collection buttons below.                                                   |
| <b>Coordinator Mode and Artist Mode</b> | The coordinator/artist mode switch. In Coordinator Mode the INFO checks are hidden, the verdict and Copy Report appear. With Coordinator Lock on (see section 9) every Fix button is hidden. |
| <b>Live</b>                             | Automatic revalidation on edits: topology, UVs, transforms, renames. New scene objects are picked up automatically; the hierarchy is rescanned on changes.                                   |

## 4.2 Score and navigation

| ELEMENT               | WHAT IT DOES                                                                                                                                                                                 |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Score block           | The validation score and a health-strip - colored dots per category (green/yellow/red by count thresholds).                                                                                  |
| Next Issue            | Cyclic navigation over the defects: selects the next problem object, frames the camera, enables its worst check. Also walks objects found by the hierarchy validator alone.                  |
| Copy Summary          | A report into the clipboard. In Artist Mode - a compact summary, signed with the validator name (see section 9) and the date.                                                                |
| Hints under the score | "Validating N/M" - queue progress; "Scene changed - re-run" - the scene changed after the run; "Settings restored - press Run" - settings were restored from the scene, a new run is needed. |

## 4.3 Validation scope

| BUTTON     | SCOPE                                                                                                            |
|------------|------------------------------------------------------------------------------------------------------------------|
| Scene      | All mesh objects of the scene                                                                                    |
| Collection | All mesh objects of the active collection (nested included)                                                      |
| Clear      | Reset the validation results                                                                                     |
| (RUN)      | The first run always takes Selected - the selection. It protects against an accidental run over the whole scene. |

## 4.4 The Pipeline Checks block

Checks are enabled here and the overlay look is configured. Categories collapse; every category has a master enable switch. Next to a check name - a colored swatch (the overlay color) and the finding counter after a run.

| ELEMENT                        | WHAT IT DOES                                                                                                                                                                                                                            |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Face Orientation   Scene Units | Face Orientation mirrors the stock Blender normal overlay (it replaces the old flipped/invalid normal counters removed from the checks). Scene Units is a separate scene-units check.                                                   |
| X-Ray                          | The overlay transparency mode. On (default) - the markers draw through walls, everything is visible "x-ray style". Off - opaque mesh walls hide the markers on the far side. The stock viewport xray (Alt+Z) always forces see-through. |
| Face Offset / Point Offset     | The offset of face fills and point markers from the surface (the cure for overlay z-fighting).                                                                                                                                          |
| Presets                        | A native dropdown of named check sets + naming rules. Icons: + save, - delete, export to file, import from file (team sharing).                                                                                                         |

## 4.5 The hierarchy block

Sits right after the NAMING category. The Scan button runs the scene structure check against pipeline conventions (see section 6): asset roots, functional layers, partitions, group suffixes. Findings are aggregated per rule - one row per rule with example objects, expanded by click. Issues only mode hides clean branches. In Live mode the scan reruns itself whenever the scene structure changes (the main scenario - accepting an asset after an FBX import).

## 4.6 The Objects list

| ELEMENT     | WHAT IT DOES                                                                                                                                                         |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Search      | An object name filter; the header counter shows "N / M".                                                                                                             |
| Issues      | The "problem objects only" switch. On by default: clean objects are hidden, the header prints "All objects clean" when there are no defects.                         |
| Object card | V/E/F/T - vertex, edge, face and triangle counters; finding rows per enabled check with counts.                                                                      |
| Sel         | Selects the defective components in edit mode and frames the camera on them (viewFit).                                                                               |
| Show        | Highlights the findings in the viewport without selecting.                                                                                                           |
| Fix         | Fix buttons on checks that can repair themselves (see sections 8.4 and 6.4).                                                                                         |
| Ign         | Ignore the check on this object: the finding leaves the counters and statuses, the check is greyed out. The ignore list is the "Ignored Issues" block under Objects. |

## 4.7 Asset Status and export

| ELEMENT                   | WHAT IT DOES                                                                                                                                                                                                                                    |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ASSET STATUS              | READY (green) / REVIEW (yellow) / CRITICAL (red, with the English captions "production-ready", "needs more work", "publish blocked") - the combined verdict over every enabled check. In Coordinator Mode hierarchy findings escalate here too. |
| Export: JSON / CSV / HTML | A full report into a file. The HTML carries a "Fix first" section - blockers by descending count with object names.                                                                                                                             |
| Pre-flight FBX / USD      | Pre-export validation: active BLOCKERs block the export with a report, WARNINGs pass with a warning.                                                                                                                                            |
| Checkpoint                | Save / Load - a snapshot and restore of the validation state (enabled checks + results). Handy before experiments: Load brings the picture back, Run recomputes.                                                                                |
| Debug Info                | The button at the very bottom: copies diagnostics into the clipboard - Blender/addon/OS versions, mode, active checks, the last 40 log lines. Attach it to bug reports.                                                                         |

## 4.8 The Coordinator panel

With Coordinator Mode on, the panel shows the verdict plate (ASSET STATUS with a caption - "Blockers must be resolved before publish" / "Warnings require artist decision"), the Copy Report button, the Critical & Warning Checks block (BLOCKER and WARNING only) and the same Export/Pre-flight/Checkpoint. Back to the artist view - the Artist Mode button in the toolbar.

## 5. CHECKERS (BLENDER)

42 checks in 7 categories + Scene Units. The set and severity match the v1.8.0 code; thresholds of some checks are configurable in Preferences (the UV tab, the Checks tab - the Hard Edges chamfer width). Update checking is described in section 9.1.

### 5.1 Scene Units (scene)

| CHECK       | SEVERITY | WHAT IT LOOKS FOR                                          | WHY IT MATTERS                                                                                                 |
|-------------|----------|------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Scene Units | BLOCKER  | System is not Metric, Unit is not Meters, Scale is not 1.0 | Wrong units break texel density, physics, FBX export (the x100 scale) and any tool relying on real-world sizes |

### 5.2 TOPOLOGY (14)

| CHECK           | SEVERITY | WHAT IT LOOKS FOR                                                                                                                                                         | WHY IT MATTERS                                                                                        |
|-----------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Non Manifold    | BLOCKER  | Edges with other than 2 adjacent faces (T-joints, edges without faces)                                                                                                    | Breaks booleans, simulations, normal computation and any export requiring a closed volume             |
| Duplicate Verts | BLOCKER  | Coincident vertices within one connected shell (a 0.01 mm threshold). Coincidences between different shells (bolts on armor) are intentional practice and are not flagged | They break normals and shading, break simulations. A product of booleans, knife and accidental Ctrl+D |
| Zero Area       | BLOCKER  | Faces with an area close to zero                                                                                                                                          | Degenerate faces produce NaN normals and break UVs. A side product of boolean and Merge               |
| Z-Fighting      | BLOCKER  | Coplanar faces overlapping each other                                                                                                                                     | Flickering in renders and real-time. The classic artifact of duplicated geometry                      |
| Lamina          | BLOCKER  | "Laminar" faces - the face contour passes the same edges twice (e.g. after extruding and folding onto itself)                                                             | Zero thickness breaks boolean operations and export                                                   |

| CHECK                | SEVERITY       | WHAT IT LOOKS FOR                                                                                                                | WHY IT MATTERS                                                                                                                                                     |
|----------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Zero Length Edges    | <b>BLOCKER</b> | Zero-length edges (including sewn face contours)                                                                                 | Degenerate geometry breaks subdivide and export                                                                                                                    |
| Ngons                | <b>WARNING</b> | Faces with 5+ edges                                                                                                              | They triangulate unpredictably in the renderer; normal artifacts on hard surfaces                                                                                  |
| Isolated Verts       | <b>WARNING</b> | Vertices not connected to any edge                                                                                               | Garbage geometry: it pollutes vertex groups and UVs, invisible in renders                                                                                          |
| Starlike             | <b>WARNING</b> | Faces invisible from their own centroid: self-intersecting contours, concavity with the centroid outside, sewn zero-length edges | Such a face triangulates incorrectly - shading and bake artifacts                                                                                                  |
| Sharp Edges Not Hard | <b>WARNING</b> | Smooth-shaded edges (dihedral angle of 30 degrees or more) between substantial faces, not marked Sharp                           | Smooth shading on a sharp corner produces a shading artifact. Skipped: custom-normal meshes, chamfer strips (threshold in Preferences - Checks), flat-shaded pairs |
| Triangles            | <b>INFO</b>    | Triangular faces                                                                                                                 | A midpoly workflow allows triangulation - artist review. Critical only in deformation zones                                                                        |
| Boundary Edges       | <b>INFO</b>    | Open edges - exactly one adjacent face                                                                                           | They can be intentional (cutouts, hidden faces). Highlighted for review                                                                                            |
| Face Aspect Ratio    | <b>INFO</b>    | Quads with an aspect ratio above the threshold (6:1 by default)                                                                  | Stretched quads produce artifacts under subdivide and skinning                                                                                                     |
| Poles                | <b>INFO</b>    | Vertices with a non-standard edge count (3, 5, >5, 0)                                                                            | Poles affect subdivide and skinning but are often intentional                                                                                                      |

### 5.3 TRANSFORMS (6)

| CHECK                 | SEVERITY | WHAT IT LOOKS FOR                                                      | WHY IT MATTERS                                                                                 |
|-----------------------|----------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Non Applied Transform | BLOCKER  | A non-zero object rotation                                             | An unapplied rotation changes normal directions on export, breaks physics and bone orientation |
| Scale                 | BLOCKER  | Scale other than 1.0 on any axis (a 0.001 tolerance)                   | A non-normalized scale distorts simulations, skeleton deformation and texel density            |
| Modifier Stack        | WARNING  | Unapplied modifiers (except Armature)                                  | A stray Subdiv, Bevel or Mirror changes the final geometry on export                           |
| Parent Geometry       | WARNING  | A mesh parented to another mesh                                        | It breaks the asset hierarchy and scripted scene assembly attempts                             |
| Origin At Zero        | INFO     | The object pivot is not at world zero                                  | It can be intentional (a loop, a hinge). Highlighted for review                                |
| Uncentered Pivots     | INFO     | The pivot is far from the center of its own bbox (>5% of the diagonal) | A convention check: buildings and locators have offset pivots - legitimate, the artist decides |

### 5.4 SYMMETRY (3)

| CHECK              | SEVERITY | WHAT IT LOOKS FOR                                      | WHY IT MATTERS                                                                                                       |
|--------------------|----------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Symmetry X / Y / Z | INFO     | Vertices without a mirror pair on the axis (a KD-tree) | Broken symmetry breaks mirror rigs and blend shapes. Enable consciously for symmetric objects - characters, vehicles |

## 5.5 UV (9)

| CHECK            | SEVERITY | WHAT IT LOOKS FOR                                                                          | WHY IT MATTERS                                                                    |
|------------------|----------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| UV Overlap       | BLOCKER  | Overlapping UV islands of different material groups                                        | Overlap = the same texels for different polygons: unique textures cannot be baked |
| UV UDIM Bounds   | BLOCKER  | UV islands crossing a UDIM tile border                                                     | An island in two tiles = the right UDIM texture cannot be assigned                |
| Uv Material Udim | BLOCKER  | Several material groups on one UDIM tile                                                   | It breaks the renderer UDIM addressing and texturing automation                   |
| UV Single Set    | WARNING  | The UV set count is not exactly 1                                                          | Several sets are a sign of unfinished work or a pipeline conflict                 |
| UV Micro Shell   | WARNING  | UV islands below the area threshold (about 6x6 px at 2048)                                 | Islands too small for texturing, wasting atlas space                              |
| UV Stretch       | WARNING  | Faces where the UV angle differs strongly from the 3D angle                                | Stretch = texture distortion: visible on patterns and the normal map              |
| UV Texel Density | INFO     | TD deviation from the target value (px/cm)                                                 | With no Target TD set - it simply shows the real value                            |
| UV Padding       | INFO     | The distance between shells below the threshold, or to the tile border below the threshold | Insufficient padding causes bleeding under mip-mapping; it can vary intentionally |
| Missing UVs      | WARNING  | Faces without UV coordinates                                                               | A texture cannot be baked or stretched onto such faces                            |

## 5.6 NAMING (6)

Naming rules are configured in Preferences - Naming (object/group prefixes and suffixes, mesh data; see section 9). The checks compare names against this policy.

| CHECK            | SEVERITY | WHAT IT LOOKS FOR                                                                                           | WHY IT MATTERS                                                                                                     |
|------------------|----------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Object Name      | WARNING  | Default names (Cube, pCube1), .001 numbering, special characters, capital letters, prefix/suffix mismatches | Broken names break asset manager scripts and exact name matching                                                   |
| Group Name       | WARNING  | The same rules for collections + the group prefix/suffix                                                    | Broken naming breaks the hierarchy, LOD systems and shot assembly                                                  |
| Mesh Data Name   | WARNING  | A mesh datablock name (Mesh.101) not matching the object name                                               | Polluted datablocks surface in scripts and on linking                                                              |
| Mat Numbering    | WARNING  | Material names with numbering (.001, .002)                                                                  | A numbered material is a copy instead of the original; it breaks the material library                              |
| Duplicated Names | BLOCKER  | Matching object names in the scene                                                                          | Name collisions break library linking and export - unlike Blender numbering this is a real conflict, not cosmetics |
| Trailing Numbers | WARNING  | Trailing numbers in names (bolt01, wheel2)                                                                  | A convention check: proper partition numbering is strictly two digits after an underscore                          |

## 5.7 MATERIALS (3)

| CHECK            | SEVERITY | WHAT IT LOOKS FOR                       | WHY IT MATTERS                                                                       |
|------------------|----------|-----------------------------------------|--------------------------------------------------------------------------------------|
| Mat Assignment   | BLOCKER  | Empty material slots                    | An object with an empty slot renders black/default - a defect in a production render |
| Missing Textures | BLOCKER  | Image Texture nodes with a missing file | A missing file = a pink/black render                                                 |
| Mat Suffix       | WARNING  | Material names without the _mat suffix  | A single suffix tells materials apart from meshes and groups in scripts              |

## 5.8 CLEANUP (1)

| CHECK       | SEVERITY | WHAT IT LOOKS FOR                                                                                                                                                 | WHY IT MATTERS                                                                                          |
|-------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Unused Data | WARNING  | Empty vertex groups, unedited shape keys, junk custom attributes (stock Blender attributes are untouched; GN attributes are protected while the modifier is live) | Junk data bloats the file and confuses auto-riggers. The Fix removes the flagged items through a dialog |

## 6. THE HIERARCHY VALIDATOR

### 6.1 Asset structure conventions

The asset hierarchy is assembled from group empties (Maya uses locators for this). The validator checks the structure against the conventions:

- the asset root - a group empty with the `_grp` suffix (truck\_a\_grp, for example);
- inside - functional layers (geo, proxy, rig, wip etc.): the set is free, with a built-in whitelist of 24 common names, extendable in Preferences;
- partitions of repeated elements - strictly two-digit numbering after an underscore: bolt\_01, bolt\_02 (bolt\_1 and bolt\_002 are violations);
- every group with a group role carries the `_grp` suffix;
- names are lowercase, no .001, no forbidden characters, no default DCC names;
- lights and cameras are not part of the asset structure and are not validated; a static asset may be grouped right at the root.

### 6.2 Rules (13)

| RULE                     | SEVERITY | WHAT IT CATCHES                                                                               |
|--------------------------|----------|-----------------------------------------------------------------------------------------------|
| Blender numbering (.001) | ERROR    | Copies with .001 auto-numbering - a sign of duplication instead of renaming                   |
| Missing group suffix     | ERROR    | A group with a group role missing the <code>_grp</code> suffix                                |
| Forbidden characters     | ERROR    | Forbidden characters in node names                                                            |
| Default DCC name         | ERROR    | Default package names (Cube, Empty, pCube1...)                                                |
| Empty group              | ERROR    | An empty with a group role and no children - an unfilled locator after import                 |
| Uppercase in name        | WARNING  | Capital letters in the name                                                                   |
| Unknown functional layer | WARNING  | A layer outside the whitelist (extended in Preferences)                                       |
| Orphan empty             | WARNING  | An empty unreachable from any asset root                                                      |
| Orphan mesh              | WARNING  | A mesh unreachable from any asset root                                                        |
| Name mismatch in group   | WARNING  | A node name not matching the <code>&lt;base&gt; / &lt;base&gt;_NN</code> pattern of its group |
| Mesh under mesh          | WARNING  | A mesh parented to a mesh (a duplicate of the Parent Geometry check)                          |
| No asset root            | WARNING  | No asset root in the scene                                                                    |
| Multiple asset roots     | WARNING  | Several roots - check that it is intentional                                                  |

## 6.3 Scanning

- Scan - a manual scan start; the result is collapsible sections per asset root with their own ERROR/WARNING counters, plus "Not connected to any root" and scene-level finding sections.
- Issues only - show problem branches only.
- Live auto-scan: with Live enabled the scan reruns itself after structure changes (an FBX import, reparenting, renames). A "Stale" badge means the result is outdated - run Scan.
- An X on a finding - ignore the rule on that node; the "N ignored - clear" counter resets the ignores.

## 6.4 Fixes

| FIX                   | WHAT IT DOES                                                                                                                                                                    |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Add _grp              | Adds the _grp suffix to a group; name collisions are skipped with a report                                                                                                      |
| Renumber              | Brings partition numbering to the two-digit base_01 grid (legal names are untouched; the rename is two-pass through temporary names - collision-free)                           |
| Adopt Orphans         | Connects orphaned nodes to a root: one root - automatically, several - with a dialog                                                                                            |
| Create Root           | Creates an asset root; an option to adopt the orphans right away                                                                                                                |
| Create Asset Skeleton | A static asset skeleton generator: the name comes from the .blend file, layers are picked from the whitelist with checkboxes. The asset is born legal, the validator accepts it |

*Every hierarchy fix rescans the structure right after applying.*

## 6.5 The coordinator gate

In Artist Mode hierarchy findings do not affect the Asset Status - a WIP asset has the right to an unassembled structure. In Coordinator Mode hierarchy errors escalate the verdict: ERROR to CRITICAL, WARNING to REVIEW. The scan produces a verdict even for scenes without tracker-checked objects. The whitelist and suffix settings live in Preferences - Naming - Hierarchy.

# 7. VIEWPORT OVERLAYS

Defects are painted over the geometry in the check color (the swatch in the check row, the palette in Preferences - Colors):

| TYPE   | HOW IT IS DRAWN                                                                                 | EXAMPLES                                                                            |
|--------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Faces  | A semi-transparent face fill (Faces Alpha)                                                      | ngons, triangles, starlike, lamina, zero area, uv stretch, missing uvs              |
| Edges  | Lines along edges (Edges Width / Edges Alpha); transform checks draw a heavy silhouette outline | non manifold, boundary edges, sharp edges, z-fighting, non applied transform, scale |
| Points | Vertex/centroid markers (Vertex Size / Point Offset)                                            | isolated verts, duplicate verts, poles, symmetry                                    |

- X-Ray (a button in Pipeline Checks): off - the mesh walls occlude the far-side markers; on (default) - the markers show through everything. The stock viewport xray forces see-through.
- Face Offset / Point Offset - the marker offset from the surface, when the overlay flickers with the geometry.
- Sel in an object card selects the defect in edit mode; Show highlights without selecting.
- Face Orientation - the stock Blender normal overlay, invoked from the panel (blue/red = the normal direction).

## 8. MODES AND TOOLS

### 8.1 Live

Live watches the types of changes and recomputes only what was touched: topology (mesh edit, boolean, join), UV coordinates, transforms, object and mesh data renames. New scene objects join the validation automatically, the "Scene changed" badge reminds about scope extensions. The hierarchy is rescanned in Live too. For heavy scenes the revalidation runs in batches with progress on the panel.

### 8.2 Presets

A preset = a set of enabled checks + naming rules. Save with the + icon next to the dropdown, delete with -. Export/import as a file: a finished preset (say "midpoly" or "hero subdiv") can be handed to other users as a file. A preset is stored in the Blender preferences and survives addon updates.

### 8.3 Reports

| FORMAT                           | CONTENTS                                                                                                                      | DESTINATION |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Copy Summary (artist)</b>     | A compact per-category summary + the validator signature and date                                                             | Clipboard   |
| <b>Copy Report (coordinator)</b> | The VALIDATION verdict: READY/REVIEW/BLOCKED + the BLOCKERS (fix first) and WARNINGS sections - the formal acceptance verdict | Clipboard   |
| <b>JSON / CSV / HTML</b>         | The full data: objects, checks, findings, hierarchy; the HTML carries the "Fix first" section                                 | File        |

*The validator signature: Preferences - Interface - Validator (the OS login by default). It lands in every report and in Debug Info.*

### 8.4 Fixes

Checks with automatic repair: Non Applied Transform, Scale (Apply Transform), Modifier Stack (Apply), Object/Group/Mesh Data Name (rename by policy), Mat Numbering, Unused Data (a cleanup dialog), the hierarchy fixes (section 6.4). The Fix button sits in the check row of an object card. In Coordinator Mode with Coordinator Lock every Fix is hidden - the coordinator looks but does not fix.

### 8.5 Checkpoint

Checkpoint Save pins the validation state into the scene (enabled checks + results), Checkpoint Load restores it. Useful before risky edits: save a checkpoint, experiment, load it back. A restore hint reminds you to press Run to recompute.

## 8.6 Pre-flight

Pre-flight FBX / USD is the gate before export: active BLOCKERS block the export with a report; WARNINGS only pass with a note. A broken asset never reaches the pipeline this way.

## 8.7 Logs and diagnostics

- A session log is written to %TEMP%/stukach.log (about 256 KB rotation into .old).
- Debug Info into the clipboard: versions, mode, active checks, the log tail - paste it into a bug report.
- On a Blender crash the log is in %TEMP% - the first thing the author asks for.

# 9. PREFERENCES

Preferences - Add-ons - STUKACH. Tabs: Interface, Overlay, UV, Naming, Checks, Colors.

| TAB       | SETTINGS                                                                                                                                                                                                                |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interface | Start in Coordinator Mode - the panel opens right in the acceptance mode; Coordinator Lock - hide every Fix button in Coordinator Mode (the coordinator validates only); Validator - the name in the report signatures. |
| Overlay   | X-Ray Overlay - marker transparency (section 7); Edges Width, Edges/Faces Alpha; Faces/Points Offset; Vertex Size.                                                                                                      |
| UV        | Texel Density: texture size, target TD (px/cm), tolerance (Target = 0 shows the value only); UV Padding: shell and tile-border thresholds; stretch and aspect ratio thresholds.                                         |
| Naming    | Objects and Groups: prefix/suffix lists (+/-); Mesh Data: the datablock suffix (_mesh by default); Hierarchy: the group suffix (_grp) and the functional whitelist (extends the 24 built-in layers).                    |
| Checks    | Hard Edges: the chamfer width threshold. An edge hugging a strip thinner than this percentage of the object size is a chamfer and is not flagged.                                                                       |
| Colors    | The overlay color of every check, a two-column grid; applied to the viewport instantly.                                                                                                                                 |

## 9.1 Updates

The Updates block: the Check for updates button compares the installed version with the latest GitHub release (one anonymous request, the UI never blocks).

- The Check for updates daily toggle (on by default) - a silent auto-check once a day; when there is nothing new, nothing appears anywhere.
- When an update exists: the Update available: X.Y.Z line in the Preferences becomes a link to the release page, and a badge appears at the top of the STUKACH panel.
- Install the update the usual way: download the zip from the release page and install it via Install from Disk (section 2); the badge goes away afterwards.

## 10. HOW TO USE

### 10.1 For the artist

- Load a ready check-set preset (Presets - import) or work with the default set; enable Live.
- RUN on the selection; widen to Scene/Collection to cover the whole scene.
- Work down the Objects list: Sel selects the defect and frames the camera, the overlay paints it in the check color, Live recomputes after edits automatically.
- BLOCKERS must be fixed (with the Fix button where possible), WARNINGS consciously accepted or fixed, INFOs at taste. Next Issue walks the remaining defects in a cycle.
- After an FBX import the new objects join Live by themselves and the hierarchy is rescanned automatically - put the structure in order with the section 6.4 fixes.
- Before delivery switch to Coordinator Mode and make sure the verdict is not BLOCKED; before experiments Checkpoint Save (section 8.5) is handy.

### 10.2 For the coordinator

- Enable Coordinator Mode: the INFO checks hide, BLOCKER and WARNING remain.
- Run the validation over Scene (or the asset collection).
- Read the ASSET STATUS verdict: CRITICAL - there are blockers, REVIEW - there are warnings requiring a decision, READY - the asset is clean.
- Copy Report puts the formal verdict with the BLOCKERS (fix first) and WARNINGS sections into the clipboard.
- If you only validate, enable Coordinator Lock (Preferences - Interface): the Fix buttons hide; the Start in Coordinator Mode flag opens the panel right in that mode.

## 11. THE MAYA VERSION

### 11.1 The panel

The STUKACH shelf button opens a panel with the same logic as in Blender: Run, the mode toolbar (Coordinator / Live / the "... " menu with coordinator flags), the score block, Next Issue, Copy Summary, Objects, ASSET STATUS, Delivery (Export / Pre-flight / Checkpoint), Debug Info. By default the panel docks to the left of the Attribute Editor; closing is the X, reopening is the shelf button (hot-reload built in).

## 11.2 Checks: 43 + two scene checks

| CATEGORY       | CHECKS                                                                                                                                                                            | DIFFERENCES FROM BLENDER                                                                                        |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| TOPOLOGY (14)  | non manifold, boundary edges, isolated verts, duplicate verts, face aspect ratio, triangles, ngons, poles, zero area, z-fighting, hard edges, lamina, zero length edges, starlike | Hard Edges instead of Sharp Edges<br>Not Hard: smooth edges with an angle of 30 degrees or more are flagged     |
| TRANSFORMS (6) | non applied transform, scale, construction history, origin at zero, uncentered pivots, parent geometry                                                                            | Construction History (a binary check + node type parsing) instead of Modifier Stack                             |
| SYMMETRY (3)   | symmetry X / Y / Z                                                                                                                                                                | same as Blender                                                                                                 |
| UV (10)        | single set, udim ready, udim bounds, material udim, overlap, micro shell, stretch, texel density, padding, missing uvs                                                            | There is UDIM Ready - the readiness of the UV set for UDIM tiles                                                |
| NAMING (6)     | object name, group name, mat numbering, duplicated names, shape names, trailing numbers                                                                                           | Shape Names - the shape node name against the transform; no Mesh Data Name (that is a Blender concept)          |
| MATERIALS (3)  | mat suffix, mat assignment, missing textures                                                                                                                                      | same as Blender                                                                                                 |
| CLEANUP (1)    | unused data (empty deformer sets, custom attributes)                                                                                                                              | same as Blender                                                                                                 |
| Scene          | Scene Units + Empty Groups                                                                                                                                                        | Empty Groups - transforms without children: the classic empty groups left after rebuilds; removed manually only |

## 11.3 Speed: the snapshot and progressive validation

Geometry is snapshotted in two OpenMaya iterator passes; 14 geometric checks run on a single traversal (no repeated scene queries). Validation runs as a queue of small batches - the Maya interface stays alive even on heavy scenes, with progress on the panel. Objects taking longer than a second are logged.

*Massive checks (hundreds of thousands of components) have a threshold: the report carries the honest count, while the overlay/selection uses an even sample of up to 5000 - the check row is marked "5000 sampled" and Sel is disabled.*

## 11.4 The overlay

The colored defect highlight is drawn by a native Viewport 2.0 plugin (stukachDrawOverride): filled faces, edge lines, vertex markers, plus a wireframe box for transform checks. If the plugin is unavailable (another OS or Maya version), the overlay automatically falls back to display layers. The colors are configured with swatches on the panel.

## 11.5 Locator hierarchy

- The asset structure is assembled in Maya with locator-groups following the section 6.1 conventions (\_grp, base\_01, lowercase names).
- After an FBX export/import the structure is checked by the hierarchy validator of the Blender version: groups made from locators arrive with the \_grp suffix and pass the naming rules.
- There is no hierarchy validation in the Maya version - structural acceptance is done in Blender.

## 11.6 Maya version notes

- Hard Edges stays silent in zones with custom normals (Weighted Normal and similar): Maya reads adjacent edges as hard - that is a package trait, not a checker bug.
- UV UDIM Bounds may flag clean objects with UVs on the tile border (uv = 1.0) - a known limitation, solved by a light seam shift.
- After STUKACH file updates always hot-reload with a second shelf button click - Maya does not re-read modules by itself.
- The log: %TEMP%/stukach\_maya.log; Debug Info copies the diagnostics into the clipboard.
- Sel buttons on massive findings (>5000) are disabled - that is the freeze protection, the counter stays honest.

## 12. TROUBLESHOOTING

Bug reports and wishes: [github.com/abyrvalg379/STUKACH/issues](https://github.com/abyrvalg379/STUKACH/issues) (Blender),  
[github.com/abyrvalg379/STUKACH\\_Maya/issues](https://github.com/abyrvalg379/STUKACH_Maya/issues) (Maya). Attach Debug Info and the reproduction scenario.

| SYMPTOM                                          | WHAT IT IS AND WHAT TO DO                                                                                                                                                                   |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| "Settings restored - press Run to revalidate"    | A stock STUKACH message: settings were restored from the scene on file load. Press Run - it is not a foreign addon and not an error.                                                        |
| A "Scene changed" or Stale badge/hint            | The scene changed after the run/scan. Re-run Run or Scan; in Live most of it is picked up automatically.                                                                                    |
| An object does not appear in Objects             | It is outside the validation scope: the first RUN takes Selected. Widen to Scene/Collection. A search filter or Issues-only is on - check the "N / M" header.                               |
| The overlay shows through the mesh               | That is the X-Ray button in Pipeline Checks (v1.7.5): turn it off and the walls will hide the far-side markers. An enabled Alt+Z (viewport xray) also forces the overlay to be transparent. |
| Overlay markers flicker with the geometry        | Raise Face/Point Offset in Pipeline Checks.                                                                                                                                                 |
| A finding seems false                            | Check the check threshold in Preferences; ignore it precisely (Ign on the object, X on a hierarchy rule). If it is systematically false - attach Debug Info and write the author.           |
| The Fix button is unavailable / missing          | The check has no auto-repair - it is edited by hand; or you are in Coordinator Mode with Coordinator Lock (the fixes are hidden on purpose).                                                |
| A big counter with Sel disabled ("5000 sampled") | The Maya freeze-protection threshold: the count is honest, selection/overlay work on a sample. Fix in areas.                                                                                |
| Maya: the panel did not appear after an update   | Do a hot-reload with a second shelf button click; check the Plug-in Manager (stukachDrawOverride.mll). A Maya restart resolves the remaining cases.                                         |
| A Blender crash / hang                           | Do not close Blender until the log is taken: %TEMP%/stukach.log + the Debug Info button before the crash. Blender crash dumps (.crash) live in %TEMP%.                                      |